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Unit 3 - Creating a Car Backup Sensor

PDF Documents

PDF Document 1: 1737525549679-Chapter 4 - unit 3 - Creating a Car Backup Sensor.pdf

This PDF document is included in the unit materials.



UNIT 3 CREATING A CAR BACKUP SENSOR

This section will explore creating a simple alert system using the ToF(Time-of-Flight) sensor and the MODI Speaker Module. This system will help us detect when an object gets too close and sound an alarm when needed.

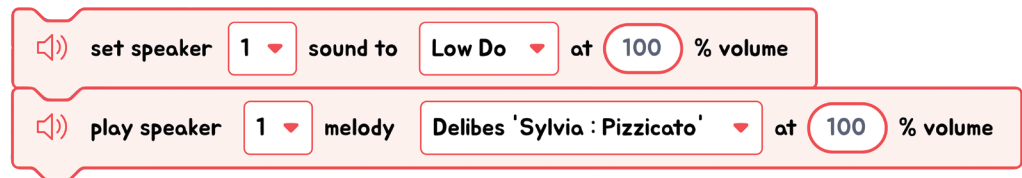


HOW IT WORKS IN ACTION:

OBJECT CLOSE When an object gets within 20 centimeters of the ToF sensor, the sensor sends a signal to the MODI system. The system instructs the speaker module to sound an alarm, which helps us know that something is too close.

OBJECT FAR If the object moves farther than 20 centimeters, the ToF sensor will no longer trigger the alarm, and the sound will turn off. This indicates that the object is now at a safe distance.

Let's look detail about the **Speaker** module.



The MODI Speaker Module is a special tool for creating and playing sounds. It's like a little speaker that can make different noises, play melodies, and even perform songs. Let's explore how this cool module works!

The MODI Speaker Module is designed to produce sound. It acts as an output module, which means it takes signals and turns them into sounds we can hear.

When connected to a MODI system, it can play different types of sounds based on your commands.

SINGLE NOTES

The speaker module has blocks that can play one note at a time. You can use these blocks to create simple melodies or play individual notes.

MUSICAL BLOCKS

Special blocks can also play pre-set tunes or songs. You need to choose the right block, and the module will play the music.





Activity: Create Restroom Indicator

1 Materials Needed:



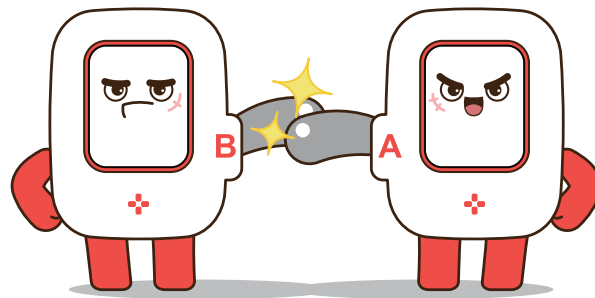
Network



ToF



Speaker



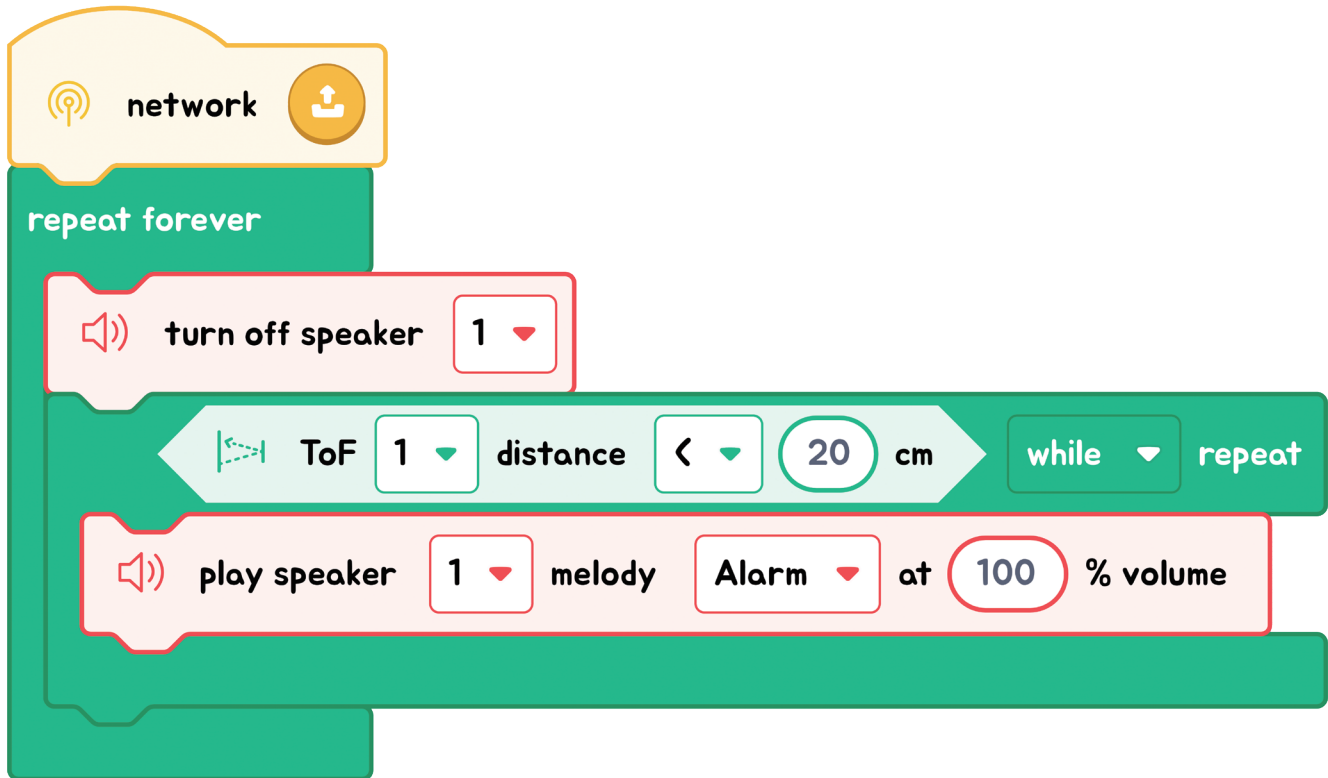
2 Code story

If Dia turn is over than **50%**

↓ true
play an **alarm**

↓ false
turn off the **speaker**

Code Quest



Checkpoint

If we want to add LED when object is near by sensor we want to show red light, otherwise the LED is off. Where should we put these blocks?

